Alberto Colombo

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Education

2017 – (December 2020) **M.Sc. Politecnico di Milano, IT** in Mathematical Engineering
Avg grade: 28.94/30
Courses: Real and functional analysis, Algorithms and parallel computing, Digital signal processing, Stochastic dynamical models, Advanced partial differential equations, Calculus of variations, Applied statistics, Model identification and data analysis, Bayesian statistics, Electronics, Plasma physics, Advanced programming for scientific computing.

2015 – 2017 **Bachelor Politecnico di Milano, IT** in Mathematical Engineering
Final grade: 110/110 cum laude
Courses: Calculus I/II/III, Linear algebra, Numerical analysis, Experimental physics I/II, Probability, Statistics I/II, Operational research, Automation and control, Economics, Chemistry, Informatics, Theoretical and applied mechanics, Thermodynamics, Partial differential equations. **Additional courses (extra credits):** Quantum physics, Data bases.

2009 – 2014 **High school - ISISS L.Geymonat - Tradate, IT**
Final grade: 100/100

Employment History

Aug. 2019 – now **Math Tutor** English Business Services, Milano - IT
Part-time trainer for the quantitative reasoning section of different graduate admission tests (GMAT, GRE, SAT).

Nov. 2018 – May 2019 **Data Science Intern** Campari S.p.A., Sesto S.Giovanni, Milano - IT
Intern in the new-born advanced analytics division. Solving business problem through data analysis and statistical modeling (demand forecasting, customer classification and profiling, autonomous CV screening, optimal sales force allocation). Communicating results to non-technical marketing/business executive decision makers.

July 2014 **Flight simulator intern** Leonardo Helicopters, Sesto Calende - IT
Short internship in the flight simulator software team. The internship was offered to the top 5 partecipants of "Valerio Filippini" physics prize.

Publications

2021 **Positioning for high performance vehicles** A. Colombo, U. Spagnolini
Submitted to: *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2021*

Projects

2020 **M.Sc. thesis** – Supervisor: U.Spagnolini
Title: *Multiple sensors augmented positioning for high performance vehicles*
Best Presentation Award at the 2019 Huawei Italy University Challenge (2° place)
Development of an accurate positioning system via bayesian sensor fusion. Application of the resulting method on high performance vehicles (i.e. F1 racing cars).

Advanced programming for scientific computing – Supervisors: L.M.Sangalli, L.Formaggia
Development of the fdaPDE Cran package (R software) in C++, concerning spatial regression with partial differential equation regularization.
The project is carried on by different groups of students of different years. Specifically, my case is about generalized linear models on 2D surfaces.

2019 **Bayesian Statistics project** – Supervisors: A.Cadonna, A.Guglielmi
Statistical analysis of spatio-temporal data (telephone traffic) on the area of Milan. Simulation via MCMC for ANOVA-DDP with sparse CAR prior.

2018 **Applied Statistics project** – Supervisors: A.Menafoglio, P.Secchi
Statistical analysis of Mappa dei rischi dei comuni italiani, particularly devoted to migration aspects within Italy. The dataset was provided by the Casa Italia department of Italian Government.

2017 **Bachelor thesis** – Supervisor: E. Lanzarone
Title: *Analysis of the predictive capabilities of electric current and pressure on the output of electrospinning process*
Developed in collaboration with IMATI division of CNR in the framework of project nano-TWICE.

Skills

Languages
Italian: Native speaker.
English: Strong reading, writing and speaking competencies. (TOEIC 2017: 845/990)

Coding
R: Expert – dplyr, shiny, ggplot2, stan.
C++: Highly proficient – eigen, expression templates, design pattern, MPI
MATLAB: Higly proficient – linear algebra, differential equations, signal processing, simulink.
Python: proficient – numpy, pandas, spark, matplotlib, sklearn.
C, Pascal: basic experience.

Other softwares
LaTeX, SQL, Linux(Ubuntu), Windows, Office.

Misc.
analytical modeling, Bayesian filtering, numerical methods for pde, teaching, training, teamworking, personal communication, public speaking.

Personal Experience

Workshops

2018 **Crash Course in Data Science**: Series of seminars on Data Science topics offered in 2018 in the passion-in-action initiative at DEIB. For a 2019 reference of the same course: Crash Course in Data Science – Passion in Action.

Theatre Experience

2017 – now **Executive Officer** – Cultural association Teatro delle Biglie
Organization and management (lectures, courses, shows,..).

Personal Experience (continued)

2019 **Assistant Teacher** – Cultural association Teatro delle Biglie
Teacher in a four months class (15 people) for performing actors.

2019 **Performer** – Gelmeroda VIII
Stage actor for an independent play held in Politecnico di Milano's auditorium.

About

Accurate, precise and methodological, can't rest until work is excellently done. My extreme curiosity led me to be a very passionate learner in many different fields, from science to technology and history. I feel comfortable working in teams and doing public speeches.

Career Aspirations

I am strongly interested in Machine Learning and Robotics. In future years I plan to start a PhD and, more in general, my long term goal is to become a researcher, either in academy or industry, in one of these fields. I am currently looking for research opportunities involving the applications of localization and mapping algorithms and Robotics in general.

Hobbies & Interests

Board-games, analog photography, hiking, biking, theatre, cognitive sciences, evolutionary biology, ancient history.